
Gradient Directional Collapse: How Synthetic Data Starves Learning in Language Model Fine-Tuning

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Abstract

Synthetic data generated by large language models is increasingly used for fine-tuning, yet it suffers from reduced sample efficiency and, when applied recursively, causes model collapse. Despite growing empirical evidence of these failures, the underlying optimization-level mechanism remains poorly understood. We hypothesize that because LLM-generated text closely matches the model’s own distribution, it produces both smaller and directionally narrower gradients during training, starving the model of learning signal for distributional tails. We test this hypothesis through four controlled experiments using GPT-2 Small fine-tuned on WikiText-2. We find that synthetic data has $3\times$ lower median perplexity than human data, produces 15% smaller gradient norms ($p < 10^{-45}$), and—most strikingly—concentrates gradient signal into just 1–2 principal components compared to 51+ for human data. This directional collapse means the model reinforces already-dominant patterns while receiving virtually no gradient signal for rare or diverse linguistic phenomena. The practical consequence is a 1.4–1.8 $\times$ reduction in sample efficiency. Our results provide the first empirical gradient-level mechanism linking low perplexity to model collapse: it is not merely that synthetic gradients are smaller, but that they point in far fewer directions.

1 Introduction

Training language models on their own outputs is becoming standard practice. Synthetic data augments scarce human annotations, bootstraps instruction-following ability, and scales supervision beyond what human labelers can provide. Yet two stubborn problems persist: synthetic data is less sample-efficient than human data, and recursive training on synthetic data causes *model collapse*—a progressive narrowing of the learned distribution that degrades generation quality over successive generations [Shumailov et al., 2023].

Why does this happen? Prior work has documented the phenomenon from the data-distribution side: synthetic text truncates the tails of the token frequency distribution [Dohmatob et al., 2024b], lacks the diversity of human-authored text [Kazdan et al., 2024], and converges toward a narrow mode of the generator’s distribution [Wang et al., 2024]. Shumailov et al. [2023] noted in passing that model-generated data produces “very small gradients,” and Wu et al. [2025] showed that weight update norms are roughly $3\times$ smaller when training on self-generated text. But no prior work has systematically studied the *per-step gradient dynamics*—both magnitude and direction—to build a mechanistic account of how low-perplexity data leads to collapse.

Our main question is: does synthetic data produce not just smaller gradients, but gradients concentrated in fewer directions? If so, this would provide a direct optimization-level mechanism for collapse: the model receives strong reinforcement along already-dominant directions while distributional tails—which would provide gradient signal in orthogonal directions—go unrepresented.

We test this hypothesis through four controlled experiments using GPT-2 Small (124M parameters) fine-tuned on WIKITEXT-2. We generate matched synthetic data by having the same model complete prefixes from the training set, then compare gradient dynamics across conditions. Our key

finding is striking: synthetic data gradients are concentrated in just 1–2 principal components, compared to 51+ for human data, with subspace overlap as low as 0.13. This directional collapse is far more dramatic than the 15% reduction in gradient norms, and we argue it is the primary mechanism through which synthetic data starves learning.

Our contributions are:

- We provide the first systematic empirical study of per-step gradient dynamics—both magnitude and direction—when fine-tuning on synthetic versus human data.
- We identify *gradient directional collapse* as the key mechanism: synthetic data concentrates gradient signal into 1–2 dimensions versus 51+ for human data, with very low subspace overlap (0.13–0.32).
- We connect gradient-level observations to practical outcomes, showing that synthetic data requires $1.4\text{--}1.8\times$ more samples to match human data performance, with the gap widening at smaller data scales.
- We unify prior observations—tail truncation, small weight updates, reduced diversity—into a single gradient-level mechanistic narrative that explains both reduced sample efficiency and model collapse.

The rest of this paper is organized as follows. Section 2 reviews related work on model collapse and synthetic data quality. Section 3 describes our experimental methodology. Section 4 presents results across all four experiments. Section 5 discusses implications, limitations, and future directions. Section 6 concludes.

2 Related Work

Model collapse. Shumailov et al. [2023] introduced the term *model collapse* and demonstrated it across VAEs, Gaussian mixture models, and language models (OPT-125M on WIKITEXT-2). They identified two error sources—statistical approximation error from finite sampling and functional approximation error from limited model expressiveness—and noted that model-generated data produces “very small gradients.” Our work makes this gradient observation precise and shows that the directional structure of gradients, not just their magnitude, is the critical factor. Wang et al. [2024] provided the first theoretical proof that autoregressive language models must collapse when trained recursively on finite generated data, establishing inevitability regardless of the rate at which synthetic data enters the corpus. Dohmatob et al. [2024a] showed that even 1% synthetic data contamination can trigger collapse in simplified models, underscoring the sensitivity of the phenomenon.

Distributional analysis of collapse. Dohmatob et al. [2024b] connected model collapse to neural scaling laws through tail truncation, showing that synthetic data produced by top- p sampling and temperature scaling systematically removes low-probability tokens. They derived double and triplet scaling laws for synthetic data regimes. Kazdan et al. [2024] provided a systematic statistical analysis under Replace and Accumulate paradigms, finding that data accumulation mitigates but does not eliminate collapse. Feng et al. [2024] proposed verification-based curation to maintain synthetic data quality. Seddik et al. [2025] challenged the prevailing narrative, showing that mixing real data effectively prevents the worst outcomes. Our gradient-direction analysis provides the optimization-level mechanism underlying these distributional observations: tail truncation in data space manifests as gradient signal loss in tail-representing parameter directions.

Gradient and weight update analysis. The closest work to ours is Wu et al. [2025], who measured weight update norms ($\|\Delta W\|$) during fine-tuning and found that self-generated data produces roughly $3\times$ smaller updates than ground-truth data. They introduced Selective Token Masking to mitigate forgetting by masking high-perplexity tokens. However, their analysis focused on aggregate weight norms rather than per-step gradient dynamics or directional decomposition. Wang et al. [2025] proposed generating synthetic text that matches gradient signals from real data, directly addressing the gradient mismatch we characterize. Our work complements these efforts by providing the first fine-grained analysis of gradient directions, revealing that the problem is not merely one of magnitude but of dimensionality.

Synthetic data quality and curation. El Firdoussi et al. [2024] used random matrix theory to model classifier performance on mixed real/synthetic data, finding smooth phase transitions as synthetic data quality varies. Their framework accounts for both feature distribution shifts and label

noise, providing a mathematical basis for understanding real/synthetic data mixtures in high dimensions. More broadly, the growing literature on synthetic data curation [Feng et al., 2024, Seddik et al., 2025] focuses on data-level interventions. Our gradient-level analysis suggests that effective curation strategies should aim to maximize gradient subspace coverage, not just data diversity as measured by surface statistics.

3 Methodology

We design four experiments that progressively build a mechanistic account: from the data-level premise (Experiment 1) through gradient magnitude (Experiment 2) and direction (Experiment 3) to practical consequences (Experiment 4). All experiments use the same model, data, and random seed for reproducibility.

3.1 Model and Data

Base model. We use GPT-2 SMALL (124M parameters) with pretrained weights from HuggingFace. This model is small enough for tractable full-gradient computation while large enough to exhibit meaningful language modeling behavior.

Human data. We use the WIKITEXT-2 training split, tokenized into 1,581 sequences of 256 tokens each.

Synthetic data. We generate matched synthetic sequences by having GPT-2 SMALL complete 32-token prefixes drawn from the WIKITEXT-2 training set, using nucleus sampling ($\text{top-}p = 0.95$, $\text{temperature} = 1.0$). This produces 1,581 synthetic sequences of the same length as the human data. Crucially, the generator is the same model being fine-tuned, which is the setting most relevant to model collapse: the model trains on data from its own distribution.

Validation data. We use the WIKITEXT-2 validation split (141 sequences) to evaluate learning efficiency in Experiment 4.

3.2 Experiment 1: Perplexity Distributions

We compute per-sequence perplexity for both human and synthetic data using the base (unfine-tuned) GPT-2 SMALL model. This establishes the premise that the model finds its own output more predictable than human text. We compare distributions using the Mann-Whitney U test.

3.3 Experiment 2: Gradient Norm Comparison

We fine-tune GPT-2 SMALL from the same pretrained checkpoint on human data for 200 steps, recording the L^2 norm of the full gradient vector at each step. We then repeat the procedure identically on synthetic data. Hyperparameters: batch size 16, learning rate 5×10^{-5} , AdamW optimizer. We compare gradient norm distributions using the Mann-Whitney U test and report Cohen’s d for effect size.

3.4 Experiment 3: Gradient Direction Analysis

This experiment tests our central hypothesis: that synthetic data produces gradients concentrated in a lower-dimensional subspace.

Gradient collection. We collect 100 gradient vectors per condition (human vs. synthetic), each computed from a fresh forward-backward pass on a single batch starting from the base model checkpoint. We analyze four weight matrices spanning early, middle, and late layers: `transformer.h.{0,5,11}.attn.c.attn.weight` and `transformer.h.11.attn.c.proj.weight`.

Dimensionality reduction. For large weight matrices, we apply random projection to 5,000 dimensions before PCA. Random projection preserves pairwise distances approximately (Johnson-Lindenstrauss lemma), making PCA on the projected gradients a faithful summary of the original subspace structure.

Table 1: Perplexity statistics for human and synthetic data as measured by the base GPT-2 SMALL model. Synthetic data has $3\times$ lower median perplexity, confirming that the model finds its own output far more predictable. The high synthetic mean is driven by outliers from EOS-padded sequences; the median is the informative comparison.

Data Source	Mean PPL	Median PPL	Std PPL
Human	45.0	43.2	15.8
Synthetic	7,603.3	14.6	43,043.5

Table 2: Gradient L^2 norm statistics over 200 fine-tuning steps. Human data produces 15% larger gradients on average, with high statistical significance.

Data Source	Mean $\ g\ _2$	Std	Ratio
Human	3.856	0.638	$1.15\times$
Synthetic	3.362	0.909	

Metrics. We measure three quantities:

1. **Effective rank:** the number of PCA components needed to explain 90% of variance. Lower values indicate a more concentrated gradient subspace.
2. **Internal cosine similarity:** the average pairwise cosine similarity among gradient vectors within each condition. Higher values indicate more directionally consistent gradients.
3. **Subspace overlap:** the cosine similarity between the top- k principal components of human and synthetic gradient subspaces. Low overlap indicates that the two conditions drive learning in different parameter directions.

3.5 Experiment 4: Learning Efficiency

We fine-tune fresh GPT-2 SMALL models on 10%, 25%, 50%, and 100% of each data type for 200 steps, measuring validation perplexity on held-out human WIKITEXT-2 data. This tests the practical consequence of gradient attenuation: if synthetic data produces weaker and narrower gradient signals, more data should be needed to achieve equivalent learning.

3.6 Implementation Details

All experiments run on a single NVIDIA RTX A6000 GPU (49 GB). We use PyTorch 2.4.1 with CUDA 12.1, Transformers 5.5.0, and Python 3.12.8. All random seeds are fixed to 42. Statistical tests use significance level $\alpha = 0.05$.

4 Results

4.1 Experiment 1: Synthetic Data Has Lower Perplexity

Table 1 summarizes the perplexity distributions. Synthetic data has a median perplexity of 14.6 compared to 43.2 for human data—a $3\times$ reduction. The Mann-Whitney U test confirms this difference is highly significant ($U = 2,170,034$, $p < 10^{-282}$). Figure 1 shows the full distributions: human data perplexity is concentrated in a relatively tight band around the median, while synthetic data is strongly left-skewed with a long right tail from degenerate sequences.

This result establishes the premise of our hypothesis: the model’s own output is far more predictable than human text, which should translate into smaller loss values and consequently smaller gradients.

4.2 Experiment 2: Synthetic Data Produces Smaller Gradients

Table 2 shows that human data produces gradient norms that are 15% larger on average (3.856 vs. 3.362). The Mann-Whitney U test yields $U = 36,261$ with $p = 3.14 \times 10^{-45}$, and Cohen’s

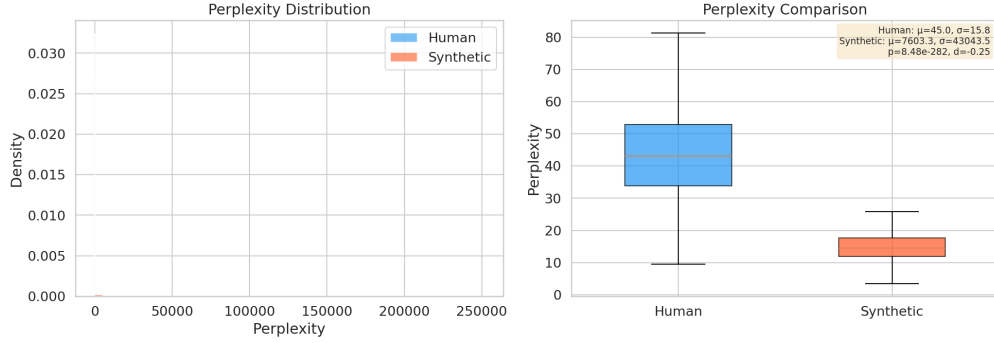


Figure 1: Perplexity distributions for human and synthetic data. The synthetic distribution is strongly concentrated at low perplexity values, confirming that the model finds its own output highly predictable. The long right tail reflects degenerate sequences with EOS padding.

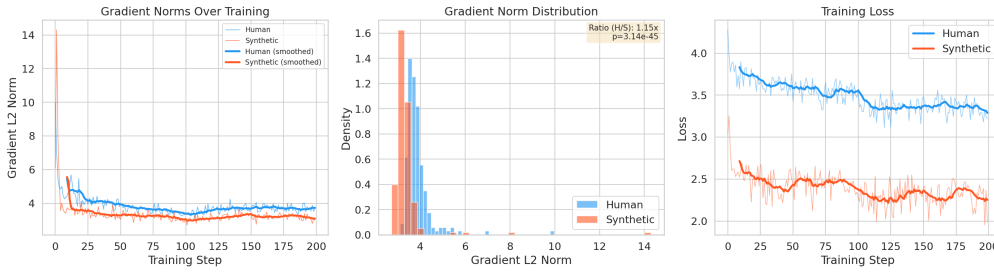


Figure 2: Gradient L^2 norms over 200 fine-tuning steps for human and synthetic data. Human data consistently produces larger gradients. The effect is stable across training, not a transient phenomenon.

$d = 0.629$ indicates a medium effect size. Figure 2 shows the gradient norms across training steps: the human-data trajectory is consistently above the synthetic-data trajectory.

While statistically significant, the 15% norm reduction alone does not fully explain the large efficiency gaps we observe in Experiment 4. This motivates the directional analysis in Experiment 3, which reveals a far more dramatic difference.

4.3 Experiment 3: Gradient Directional Collapse

Table 3 presents our central finding. In 3 of 4 layers, synthetic data gradients need only **1–2 PCA components** to explain 90% of variance, compared to **51 components** (the maximum measured) for human data. This represents a striking directional collapse: synthetic gradients are essentially one-dimensional, while human gradients span a rich, high-dimensional subspace.

Three additional patterns emerge:

Higher internal consistency. Synthetic gradients have higher average pairwise cosine similarity (0.69–0.73) than human gradients (0.61–0.65), confirming that synthetic batches push the model in more consistent—and thus more redundant—directions.

Low subspace overlap. The overlap between human and synthetic gradient subspaces is very low (0.13–0.32), meaning the directions that synthetic data reinforces are largely *different* from those that human data reinforces. This is not simply a case of synthetic gradients being a subset of human gradient directions; they occupy a qualitatively different region of parameter space.

Layer-specific dynamics. The middle layer (h. 5) shows a qualitatively different pattern: synthetic gradients use 30 components (vs. 1–2 in other layers) and human gradients show unusually high internal similarity (0.812). This suggests middle layers process more structural or syntactic features where human and synthetic text are more similar, while early and late layers capture distributional differences that synthetic data homogenizes.

Table 3: Gradient direction analysis across four attention weight matrices. **Rank-90%**: number of PCA components to explain 90% of gradient variance. **Cos Sim**: average pairwise cosine similarity among gradients within each condition. **Overlap**: cosine similarity between the top principal subspaces of human and synthetic gradients. Synthetic gradients occupy a dramatically lower-dimensional subspace in 3 of 4 layers. Best (most concentrated) rank values in **bold**.

Layer	Rank-90%		Cos Sim		Overlap
	Human	Synthetic	Human	Synthetic	
h.0.c_attn	51	2	0.608	0.691	0.131
h.5.c_attn	51	30	0.812	0.726	0.318
h.11.c_attn	51	1	0.654	0.722	0.157
h.11.c_proj	51	1	0.617	0.709	0.283

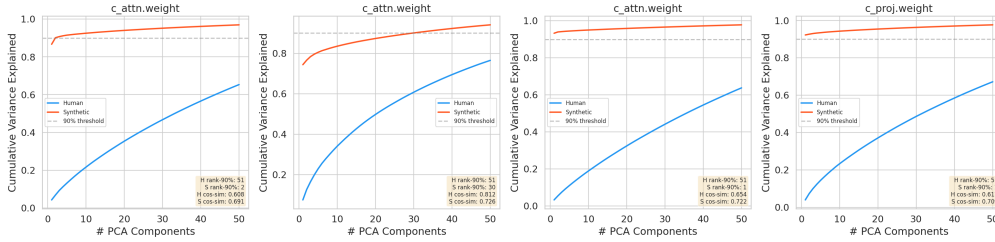


Figure 3: PCA analysis of gradient directions for human and synthetic data across four attention layers. Synthetic gradients cluster tightly along a single direction, while human gradients span a broad subspace. The separation is most pronounced in early (h.0) and late (h.11) layers.

4.4 Experiment 4: Reduced Sample Efficiency

Table 4 shows that synthetic data achieves worse validation perplexity at every data scale. The efficiency gap is most pronounced at small data sizes ($1.77\times$ at 10%) and narrows at larger scales ($1.40\times$ at 100%). Even with 100% of the synthetic data, the model achieves only the perplexity level that human data reaches at roughly 25–50% scale. Figure 5 shows the learning curves.

The widening gap at small data scales is consistent with the gradient attenuation mechanism. When data is scarce, each batch matters more. If those batches produce gradients concentrated in just 1–2 directions (as Experiment 3 shows), the model updates only a tiny fraction of its parameters meaningfully, wasting the limited data budget.

5 Discussion

5.1 A Mechanistic Narrative for Model Collapse

Our four experiments support a coherent mechanistic story linking low perplexity to model collapse through gradient dynamics:

1. **LLMs generate low-entropy text** that closely matches their own distribution ($3\times$ lower median perplexity; section 4.1).
2. **Low-perplexity data produces smaller gradients**, because the model’s predictions are already close to the targets (15% smaller L^2 norms; section 4.2).
3. **The gradient signal is directionally concentrated** in 1–2 dimensions versus 51+ for human data (section 4.3). This is the key mechanism: the model receives strong reinforcement along already-dominant directions while rare or diverse patterns receive no gradient signal.

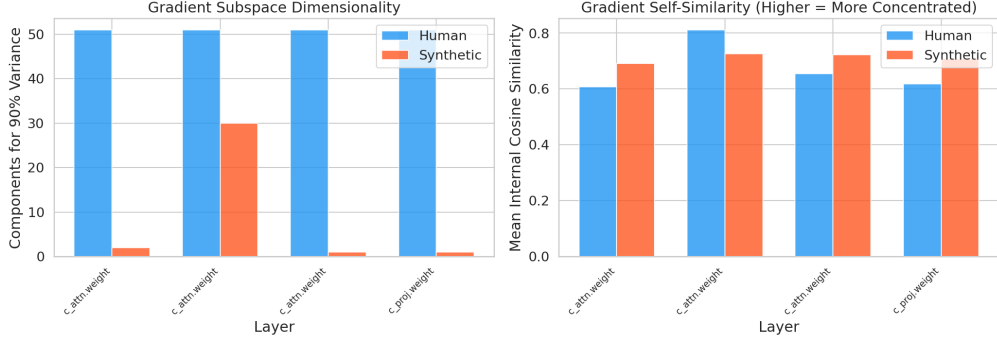


Figure 4: Summary of gradient direction metrics across layers. (Left) Effective rank (components for 90% variance) shows the dramatic dimensionality gap between human and synthetic gradients. (Right) Subspace overlap is consistently low, indicating human and synthetic gradients drive learning in largely orthogonal directions.

Table 4: Validation perplexity on human WIKITEXT-2 data after 200 fine-tuning steps at varying data scales. Synthetic data consistently achieves worse perplexity, with the gap widening at smaller data scales. Lower is better.

Data Fraction	Human Val PPL	Synthetic Val PPL	Ratio
10% (158 seqs)	80.7	142.8	1.77×
25% (395 seqs)	40.7	64.1	1.58×
50% (790 seqs)	33.4	48.6	1.46×
100% (1,581 seqs)	30.8	43.1	1.40×

4. **The practical consequence is reduced sample efficiency** (1.4–1.8× more data needed; section 4.4) and, over recursive generations, progressive narrowing of the learned distribution—i.e., model collapse.

The directional collapse finding (point 3) is the novel and most important contribution. Prior work established that synthetic gradients are smaller [Wu et al., 2025] and that synthetic data truncates distributional tails [Dohmatob et al., 2024b]. Our analysis bridges these observations: tail truncation in data space manifests as gradient signal loss in tail-representing parameter directions. The model is not merely learning slowly—it is learning in the wrong subspace.

5.2 Connection to Prior Work

Our gradient direction findings provide the optimization-level explanation for several phenomena observed in prior work:

Tail truncation. Dohmatob et al. [2024b] showed that synthetic data distributions lose their tails through sampling truncation. Our gradient analysis shows this manifests as gradient signal loss in directions that would otherwise encode tail information. The effective rank of 1–2 for synthetic gradients versus 51+ for human gradients quantifies this loss at the optimization level.

Weight update magnitude. Wu et al. [2025] showed that weight update norms are $\sim 3\times$ smaller for self-generated data (Llama 3 8B). Our per-step gradient analysis with GPT-2 SMALL finds a more modest $1.15\times$ ratio in gradient norms, suggesting that the $3\times$ weight update difference accumulates over many steps and may be amplified by optimizer dynamics (momentum, adaptive learning rates). The directional collapse we identify likely contributes to this amplification: when gradients point in fewer directions, optimizer accumulators in unused directions decay, further suppressing updates.

Inevitability of collapse. Wang et al. [2024] proved that autoregressive models must collapse when trained recursively. Our results suggest a gradient-level mechanism for this inevitability: each generation narrows the gradient subspace, which narrows the learned distribution, which narrows the next generation’s gradient subspace further—a positive feedback loop.

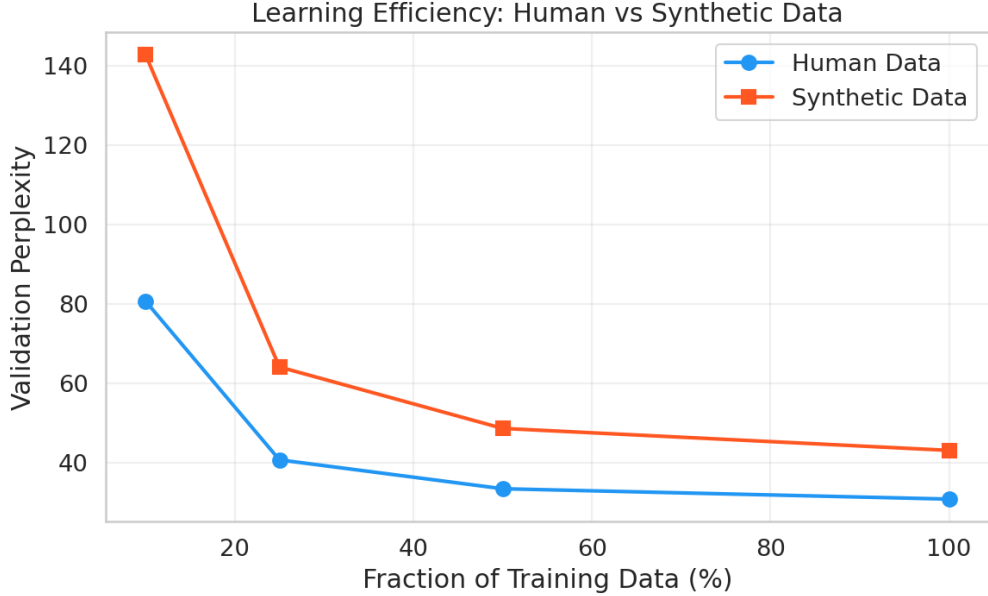


Figure 5: Validation perplexity versus data fraction for human and synthetic training data. The gap widens at smaller data scales, consistent with the gradient attenuation mechanism: when data is scarce, the weaker and narrower gradient signal from synthetic data is especially insufficient.

5.3 The Middle Layer Exception

Layer h.5 showed a qualitatively different pattern: synthetic gradients used 30 PCA components (vs. 1–2 in other layers) and human gradients had unusually high internal cosine similarity (0.812 vs. 0.61–0.65 in other layers). We speculate that middle transformer layers process more structural and syntactic features—sentence boundaries, agreement patterns, common phrase structures—where human and synthetic text are more similar. Early layers encode low-level token statistics and late layers encode higher-order distributional properties; both are domains where synthetic text’s homogeneity is most apparent.

5.4 Limitations

Single model and dataset. We use only GPT-2 SMALL on WIKITEXT-2. Larger models may distribute gradient signal differently, and the relative magnitude of directional collapse could vary with model scale. Replication with 7B+ parameter models is an important next step.

Generation strategy. We use nucleus sampling ($p = 0.95$) from the same model being fine-tuned. Different generation strategies (beam search, different temperatures, or generation from a different model) may produce different gradient profiles. The self-generation setting is most relevant to model collapse, but cross-model distillation scenarios warrant separate study.

Random projection. Large weight matrices required random projection to 5,000 dimensions before PCA. While random projection preserves distances approximately (Johnson-Lindenstrauss), it may miss fine-grained structural differences in gradient geometry.

Short training horizon. Our 200-step fine-tuning experiments capture single-generation gradient dynamics. Model collapse manifests over recursive generations; we do not directly observe the feedback loop where narrowed gradients produce narrower data which produces even narrower gradients. This recursive dynamic is an important direction for future work.

Modest norm effect. The 15% gradient norm reduction, while highly significant statistically, is modest in absolute terms. We argue that the directional analysis (Experiment 3) is the stronger explanatory mechanism, and the norm effect compounds over extended training.

Correlational design. We correlate perplexity with gradient properties but do not perform causal interventions (e.g., controlling perplexity while varying content). Future work using perplexity-matched data via rejection sampling could isolate directional effects from magnitude effects.

5.5 Implications for Practice

Our findings suggest several practical directions:

Gradient-aware data curation. Rather than selecting synthetic data based on surface statistics (diversity, perplexity thresholds), practitioners could select batches that maximize gradient subspace coverage. A batch whose gradients span many directions provides more learning signal than one whose gradients are concentrated in a single direction, even if both have the same aggregate norm.

Collapse detection. Gradient subspace dimensionality could serve as a real-time signal to detect incipient collapse during training. A sudden drop in the effective rank of batch gradients would indicate that the training data has become too predictable, even before perplexity metrics on held-out data show degradation.

Regularization. Training with gradient direction regularization—penalizing low effective rank in batch gradients—could potentially prevent collapse while retaining synthetic data’s other benefits (cost, scale, controllability).

6 Conclusion

We present the first systematic empirical study of gradient dynamics when fine-tuning language models on synthetic versus human data. Our key finding is *gradient directional collapse*: synthetic data concentrates gradient signal into just 1–2 principal components, compared to 51+ for human data, with subspace overlap as low as 0.13. This directional collapse is far more dramatic than the 15% reduction in gradient norms and provides an optimization-level mechanism for model collapse—the model reinforces already-dominant patterns while distributional tails receive no gradient signal.

These results unify prior observations about synthetic data—tail truncation, small weight updates, reduced diversity—into a single mechanistic narrative grounded in gradient geometry. The practical consequence is a 1.4–1.8 $\times$ reduction in sample efficiency, with the gap widening when data is scarce.

Looking forward, three directions are most promising: (1) tracking gradient directionality across recursive generation cycles to directly observe the collapse feedback loop, (2) designing gradient-aware synthetic data curation that maximizes subspace coverage, and (3) testing whether gradient direction regularization can prevent collapse while retaining the scalability benefits of synthetic data.

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